

Testing of Wakeup on CHANGE Interrupt

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Summary.

A test was performed to see if it is possible to wake up an ATmega328P microcontroller from deep sleep using a CHANGE mode on the external interrupt 0, as opposed to the FALLING mode in our current low power LoRa sensor. If successful, additional low power LoRa sensor applications are possible, e.g. statusing a remote door or gate.

The results of this test clearly prove that this works! This means that these new applications can be built using our existing hardware design and PCBs and requiring only a few minor changes to the microcontroller software.

Problem.

The current LoRa sensor firmware puts the ATmega328P microcontroller into deep sleep to conserve battery power. A *FALLING* edge signal on External Interrupt 0 causes the microcontroller to wake up and send out a LoRa message. The microcontroller waits for a reply to the message from the Hub and then goes back into deep sleep.

A jumper on the PCB allows either a high-going or low-going logic signal to generate the falling edge and thus wakeup the microcontroller.

Triggering wakeup on a single transition works for applications such as mailbox sensors and help buttons, as we only want one action to be recorded and logged. However, other applications may require both a low-high transition and a high-low transition to wake up the microcontroller and to send out a LoRa message. Examples of applications that might require this are a remote gate or door that we wish to know if it is open or closed. In these types of applications, the microcontroller should wakeup on both RISING and FALLING logic levels. When awake, the microcontroller can read the status of the interrupt pin (as a digital GPIO) to determine the current new state (open or closed) and to send out a LoRa message that includes this information. When the LoRa messages are logged, the latest message (door opened or door closed) will indicate the current state without the microcontroller needing to be woken up any extra times.

Solution to be Tested.

The simplest solution would be to change the interrupt to wake up the microcontroller on BOTH FALLING and RISING edges of the interrupt signal. Assuming that this is possible, the firmware would be modified to read the interrupt pin after awakening and to include the pin status in the LoRa message. If this can be done, the current hardware (PCB) need not be changed – only a small change to the firmware would be required.

The `attachInterrupt()` function call on line 333 in the Arduino firmware takes a MODE attribute that determines what action on the external interrupt pin will cause an interrupt. The attribute FALLING is used in the currently released code, which (in our case) actually wakes up the microcontroller on a FALLING edge on this pin (interrupt0, pin 2). Arduino also supports the attribute CHANGE that interrupts the microcontroller on both a rising and a falling edge. The question for this test is whether CHANGE will wakeup the microcontroller from deep sleep on both edges, as opposed to interrupting it when it is awake and running code. The ATmega329P documentation is unclear on this point, as the original datasheet said that edge triggered interrupts would not wakeup the microcontroller, which was found to be untrue. As of this date, the datasheet was updated to say that a FALLING edge on an external interrupt pin would wake up the microcontroller, but does not address a CHANGE interrupt.

Test Performed.

Today, I created a new branch in the “LoRa experiments” GitHub repo called “changeInterrupt”. I modified the code “RangeTestSensor.ino” in the “arduinoSketchbook” folder to comment out line 333 and replace it with:

```
attachInterrupt(digitalPinToInterrupt(BUTTON_PIN), ISR_wakeAndSend, CHANGE);
```

I also designated this code version 2.10 T (T for Test) in the up-front comments.

I recompiled this software and flashed it to the ATmega328P microcontroller on my test and development LoRa sensor board. I then tested by:

- Pressing the Test button and holding it pressed.
- Observed the LED flashes and Particle event generated to verify that the previous functionality still worked.
- I then waited a few seconds and released the Test button.
- I observed that the microcontroller again woke up and flashed the LEDs and generated the Particle event.

Prior to making this change, I verified that LED flashes and Particle events only were generated when the Test button was pressed and were not generated when the Test button was subsequently released.

I repeated this test several times to verify the operation.

I uploaded this code to GitHub (my repo only) in the “changeInterrupt” branch, leaving the released version 2.0 code in the “main” branch of this repo.

Conclusion.

The conclusion is that we can wake up the microcontroller on both RISING and FALLING edges (e.g. both gate opening and closing), and thereby track the state of a gate or door. We can do this with a simple software change to RangeTestSensor.ino:

- Change attachInterrupt() in line 333 to designate the MODE as CHANGE in lieu of FALLING. (this was tested)
- Add a line of code somewhere after the microcontroller wakes up and before the LoRa message is built and sent out that reads the interrupt0 GPIO pin to determine the current state (high or low) of the sensor. (This was not tested).
- Modify the transmitted LoRa message accordingly. (This was not tested).

It is therefore possible to expand the range of applications of our low power LoRa sensor without making any hardware changes – using our existing PCB for these new applications.

NOTE. The software changes that were not tested are expected to work with no problems or issues but the details would be application specific and this test is not for any specific new application requirement.